Ternary derivative controls and Reed-Muller weight caps: C44 < 2.1974348818

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4 November 2026; bit-convention revision 9 November 2026

Candidate, unrefereed. This result was found by AI agents working under my direction. I have only reviewed it at summary level, and other AI models reviewed it, so no human has checked it line by line.
The verifiers check the arithmetic, not the written lemmas.

Statement. *For the constant in* $r(f) \le C\, 2^{\deg(f)}$,

$$C_{44} \le \frac{1740987279236787915969864029 33}{7922816251426433759354395 0336} < 2.1974348818.$$

The lower bound is 1.5.

Editorial remark. The comparison with the exact rational is non-strict, as in the frozen proof's equation (12). The source README states it as strict; no strictness argument at the rational endpoint is supplied here. Since the rational is strictly below 2.1974348818, the decimal bound $C_{44} < 2.1974348818$ holds either way.

The seven numbered sections below follow the frozen `PROOF.md`, which uses the ± 1 convention. On 9 November 2026 the write-up was converted to the $\{0, 1\}$ convention, and Section 4 was rewritten and shortened following Alexandr Valyuzhenich's suggestion, with all conclusions unchanged. Marked corrections reproduce the supplied errata without silently changing the original argument. Supplemental material follows Section 7.

Frozen source title: Ternary derivative controls and higher weight caps

Candidate, unrefereed. This is a written mathematical proof with exact arithmetic certificates, not proof-assistant formalization or independent human refereeing. Two separately written implementations check the finite artifact. Their finite controls do not prove the named external theorems. No novelty or priority is claimed. The framework is due to CHS/Wellens; the low binary weights are due to Berlekamp–Sloane and Kasami–Tokura.

1 Definitions, support, and extremizers

The cube $\{0,1\}^n$ has independent uniform bit coordinates, and Boolean functions take values in $\{0,1\}$. Every real function has a unique multilinear real expansion $f(x) = \sum_S c_S x^S$, where $x^S = \prod_{i\in S} x_i$ and $x^\emptyset = 1$. Its real degree is the largest $|S|$ with $c_S \neq 0$ (constants have degree zero). A coordinate is relevant precisely when some such S contains it. Its local degree d_i is the largest $|S|$ containing it. Set

$$\begin{aligned} D(f) &= \sum_{i \text{ relevant}} 2^{-d_i}, \\ R_d &= \max \text{ number of relevant coordinates at degree at most } d, \\ W_d &= \max D(f) \text{ at degree at most } d. \end{aligned}$$

A nonzero multilinear real polynomial of degree at most k is nonzero on at least a 2^{-k} fraction of the cube. Here is an induction proof. The case $k = 0$ is constant. For a nonconstant polynomial choose a coordinate j and write $P = x_j A + B$ with A nonzero and degree at most $k-1$. The induction hypothesis gives support density of A at least $2^{-(k-1)}$. For each input in that support, the two values $B + A$ and B cannot both vanish. Averaging the two j branches gives support density at least 2^{-k}. This also handles polynomials whose degree is strictly below the stated upper bound.

For a Boolean $f : \{0,1\}^n \to \{0,1\}$, write

$$A_i = f(x_i{=}1) - f(x_i{=}0),$$

where $f(x_i{=}a)$ denotes f evaluated with coordinate i set to a. For relevant i, its values are $0, +1, -1$, its real degree is $d_i - 1$, and $p_i = \Pr(A_i \neq 0) = \mathbb{E}A_i^2 = \mathrm{Inf}_i(f)$. The support lemma gives $p_i \geq 2^{1-d_i}$. For the energy bounds here and in Section 2 only, use the orthonormal Walsh characters $\chi_S(x) = \prod_{j\in S}(1-2x_j)$ and coefficients $\widehat{u}(S) = \mathbb{E}[u\chi_S]$. Orthogonality follows from independence and $\mathbb{E}(1-2x_j) = 0$, so these 2^n characters form a basis and Parseval applies. The substitutions $y_j = 1-2x_j$ and $x_j = (1-y_j)/2$ do not increase multilinear degree, in either direction; thus the largest Walsh mask size equals real degree. The same holds for local degree by applying these substitutions to the nonzero derivative in that coordinate. Taking a bit difference of χ_S gives $-2\chi_{S\setminus\{i\}}$ if $i \in S$ and zero otherwise. Hence Parseval gives

$$I(f) = \sum_i p_i = 4\sum_S |S|\widehat{f}(S)^2 \leq 4\deg(f)\,\mathrm{Var}(f) \leq \deg(f).$$

Here $\mathrm{Var}(f) = \mathbb{E}f(1-\mathbb{E}f) \leq 1/4$ because $f^2 = f$. Consequently $D(f) \leq I(f)/2 \leq d/2$ and every relevant variable has influence at least 2^{1-d}. Thus $R_d \leq d\,2^{d-1}$. After removing dummy variables and relabeling, only finitely many Boolean functions need be considered. The two displayed maxima therefore exist.

We use the CHS identity $W_d = R_d/2^d$, and include its proof because its extremal reduction matters here. Choose f maximizing D at degree at most d. Let L be its relevant coordinates of local degree below d, and Y the others. If L is nonempty, take a disjoint copy L' and a fresh bit selector z, and form

$$h = zf(x_L, Y) + (1 - z)f(x_{L'}, Y).$$

This is Boolean. All degree-d terms of the two f copies involve only Y, so those terms cancel in the coefficient of z. Hence $\deg(h) \le d$. Every Y coordinate retains local degree d. Each original L coordinate and its copy have local degree exactly one more than before: a largest multilinear term containing that coordinate gives a noncanceling term also containing z. Their two weights sum to the original weight. The selector is relevant because the two copies differ for some inputs whenever L is nonempty. It adds positive weight. Thus $D(h) > D(f)$, a contradiction. An extremizer for D has every relevant local degree d, proving $W_d \le R_d/2^d$. Conversely, an R_d extremizer has $D \ge R_d/2^d$. This proves the identity.

We also need every coordinate of an R_d extremizer to have local degree d. Replace a coordinate of smaller local degree by the AND of two fresh bits. The new polynomial has degree at most d, since terms using that coordinate increase degree by at most one, and other terms keep their degrees. Both fresh coordinates are relevant: fix the other fresh bit to 1 and use a witnessing pair of inputs for the original coordinate. All other relevant coordinates remain relevant, by realizing either value of the substituted AND. The relevant count increases by one, contradicting maximality. This proves the required all-local-degrees-d reduction for count extremizers separately.

The published exact result $R_3 = 10$ supplies $W_3 = 5/4$. The elementary influence bound supplies $W_1 \le 1/2$ and $W_2 \le 1$. The R_3 premise is the result for arbitrary degree-at-most-three Boolean functions, not a restricted coloring class; see Tarannikov–Kirienko, ePrint 2000/050, and Krotov–Valyuzhenich, `https://arxiv.org/html/2311.05566v2` , section 3. CHS's identity is Corollary 2.2 in the final Combinatorica paper, `https://par.nsf.gov/servlets/purl/10201042` ; the corrected duplication proof also appears in Chiarelli's thesis, Proposition 8.5 and Corollary 8.6, `https://sites.math.rutgers.edu/~zeilberg/Theses/JohnChiarelli.pdf` .

2 A general control bound for ternary polynomials

Let A take values $0, +1, -1$, have real degree at most $k \ge 1$, and have h relevant coordinates. Put $p = \mathbb{E}A^2$. For a relevant coordinate j,

$$\Delta_j A = \frac{A(x_j{=}1) - A(x_j{=}0)}{2}$$

is a nonzero real polynomial of degree at most $k - 1$. Its nonzero values belong to $\{+1/2, -1/2, +1, -1\}$, so their squared magnitude is at least $1/4$. By the support lemma,

$$\mathbb{E}(\Delta_j A)^2 \ge \frac{1}{4}\, 2^{-(k-1)} = 2^{-(k+1)}.$$

For the Walsh characters defined in Section 1, $\Delta_j \chi_S$ is $-\chi_{S\setminus\{j\}}$ if $j \in S$ and zero otherwise. Orthogonality, first for each half-difference and then summed over coordinates, gives

$$\sum_j \mathbb{E}(\Delta_j A)^2 = \sum_S |S| \widehat{A}(S)^2 \le k \sum_S \widehat{A}(S)^2 = kp.$$

There are h nonzero summands, each at least $2^{-(k+1)}$. Therefore

$$h \le \left\lfloor kp\,2^{k+1} \right\rfloor. \tag{1}$$

For the derivative A_i of a degree-d Boolean function set $k = d-1$, $M = 2^k$ and $\lambda = Mp_i$. Formula (1) becomes

$$h_i \le \lfloor 2k\lambda \rfloor. \tag{2}$$

No affine change of coordinates over $\mathbb{F}_2$ is being used to infer real polynomial degree. Formula (1) is a direct real Fourier calculation. At $\lambda = 1$ it recovers the worst $2k$ budget in the minimum-support classification of Valyuzhenich, Theorem 2(2), `https://arxiv.org/html/2303.10995` . That external classification is useful context but is not needed as a premise of (1).

3 Restrictions controlled by any nonzero derivative

For completeness, the weighted restriction inequality for a coordinate set H is

$$D(f) \le \sum_{j \in H} 2^{-d_j(f)} + \mathbb{E}_\alpha D(f_\alpha). \tag{3}$$

To prove it for one fixed coordinate j, consider another relevant coordinate i. If i survives in both branches, its local degrees cannot increase, so both branch weights are at least its original weight. If it survives in exactly one branch, its derivative factors as x_j or $1 - x_j$ times the nonzero branch derivative. Its original local degree is exactly one larger than its surviving branch local degree. Half that branch weight therefore equals the original weight. A relevant i cannot disappear in both branches. Sum these inequalities over i other than j. Applying them successively to coordinates in H, or equivalently inducting on $|H|$, proves (3) for the coordinates outside H; the original weights of H supply the first term. Disappearing coordinates contribute zero on their branches.

Correction: Read the preceding iteration as follows: iterate over H for each coordinate outside H, then charge H at its original weights. Charging each later coordinate at its weight after restriction gives an inequality in the wrong direction. This is correction 1 in `CORRECTIONS.md`, reproduced below.

Choose a count extremizer f with $N = R_d$ and all local degrees d. Choose one of its coordinates i and let H be the full relevant-coordinate set of A_i; write $h = |H|$. Decompose $f = x_i A_i(x_H) + B(x_H, x_Y)$, where $B = f(x_i{=}0)$. On a control input α with $A_i(\alpha) = 1$, the two bit-valued branches are B and $B + 1$, forcing $B = 0$ for every Y. If $A_i(\alpha) = -1$, they are B and $B - 1$, forcing $B = 1$ for every Y. Thus $B = (1 - A_i(\alpha))/2$ on either active control input, and f_α is x_i or $1 - x_i$, with $D = 1/2$. The fraction of active control assignments is exactly p_i.

Every degree-d monomial meets H for $d \ge 2$. For a degree-d monomial in B disjoint from H, its coefficient cannot change on restricting H, since every larger monomial containing it would have degree above d. It is a nonconstant monomial in Y, but on any active control assignment B is constant in Y, so that coefficient must vanish. This is a contradiction. A degree-d monomial involving i contains a nonempty mask of A_i (of size $d - 1$), hence also meets H for $d \ge 2$. Consequently every f_α has degree at most $d - 1$.

Let $c \ge W_{d-1}$ with $c \ge 1/2$. Formula (3), all original local degrees being d, and the dictator branches imply

$$\frac{N}{2^d} \le \frac{h}{2^d} + \frac{p_i}{2} + (1-p_i)c.$$

Use (2) and $\lambda = Mp_i$ to obtain the essential inequality

$$\begin{aligned} N &\le F_d(c,\lambda) \\ &:= 2^d c + \lfloor 2(d-1)\lambda \rfloor + (1-2c)\lambda. \end{aligned} \tag{4}$$

Choose i of minimum influence. Since $\sum_i p_i \le d$,

$$1 \le \lambda \le \frac{dM}{N}. \tag{5}$$

Equations (4),(5) apply to the SAME chosen derivative. They do not assume that every low-weight derivative has the same normal form.

4 Exact energy grid and the higher low-weight spectrum

Editorial remark. Rewritten 9 November 2026 in the $\{0,1\}$ convention (see the note at the start). The conclusions are unchanged. The previous version is at commit `206a6c7` of `rpinho/math-certificates`.

Section 3 bounds N in terms of the real parameter $\lambda \ge 1$ of one chosen derivative. This section shows that λ can take only special values. Equation (6) puts λ on the grid $(2/M)\mathbb{Z}$. The Reed–Muller weight theorem then limits the values below 2 to the finite list (8). Section 5 checks the values in (8) below 2 and the necessary grid representatives (9) at and above 2.

Throughout this section, $f : \{0,1\}^n \to \{0,1\}$ has real degree at most $d \ge 2$, i is relevant, $k = d-1 \ge 1$, $M = 2^k = 2^{d-1}$ and $\lambda = Mp_i$.

Integer coefficients. Write $f(x) = \sum_T c_T x^T$. If 1_U is the bit vector equal to one exactly on U, then $f(1_U) = \sum_{T \subseteq U} c_T$. Consequently

$$c_T = \sum_{U \subseteq T} (-1)^{|T|-|U|} f(1_U) \in \mathbb{Z}.$$

Indeed, substituting the preceding identity into this sum gives $\sum_{V \subseteq T} c_V \sum_{V \subseteq U \subseteq T} (-1)^{|T|-|U|}$; the inner sum is $(1-1)^{|T \setminus V|}$, hence one for $V = T$ and zero otherwise. The bit difference therefore has the expansion

$$A_i = \sum_{T \ni i} c_T x^{T \setminus \{i\}} = \sum_T a_T x^T, \qquad a_T \in \mathbb{Z}, \quad a_T = 0 \text{ if } |T| > k.$$

Here and below the masks in the final sum omit i; the bound follows because $c_T = 0$ for $|T| > d$.

Energy grid. For uniform bits, $(x^T)(x^U) = x^{T \cup U}$ and $\mathbb{E}x^{T \cup U} = 2^{-|T \cup U|}$, by independence. Thus

$$p_i = \mathbb{E}A_i^2 = \sum_T a_T^2 \, 2^{-|T|} + 2 \sum_{T<U} a_T a_U \, 2^{-|T \cup U|},$$

where $<$ is any fixed ordering of the masks, so each distinct pair is counted once in the second sum. Every diagonal term is an integer multiple of 2^{1-2k}: its ratio to this number is $a_T^2 2^{2k-1-|T|}$,

an integer since $|T| \le k$ and $k \ge 1$. Every off-diagonal term is also such a multiple: its ratio is $a_T a_U 2^{2k-|T\cup U|}$, an integer since $|T \cup U| \le 2k$. It follows that $p_i \in 2^{1-2k}\mathbb{Z}$. Multiplying by $M = 2^k$ proves

$$\lambda \in \frac{2}{M}\mathbb{Z}. \tag{6}$$

Support word. Reduce the integer coefficients a_T modulo two. Evaluation at a bit input commutes with this reduction, so the resulting polynomial over $\mathbb{F}_2$ has degree at most k and value $A_i(x) \bmod 2$. Because A_i takes values $0, +1, -1$, this value is one exactly on its support. By definition $\mathrm{RM}(k, n)$ consists of the evaluation vectors of multilinear $\mathbb{F}_2$ polynomials of degree at most k on $\{0,1\}^n$. Thus this support indicator is a word of $\mathrm{RM}(k, n)$; it is nonzero because i is relevant. We may append unused coordinates until $n \ge k+2$: each value is repeated equally often, leaving p_i, λ and the degree bounds unchanged. For completeness the minimum distance of this code is 2^{n-k}. The support induction in Section 1 also works over $\mathbb{F}_2$: writing $P = x_j A + B$, the values B and $B + A$ cannot both vanish when $A \ne 0$. It gives weight at least 2^{n-k} for a nonzero polynomial of degree at most k; the monomial $x_1 \cdots x_k$ attains it. The support word has weight $w = 2^n \Pr(A_i \ne 0) = 2^n p_i$, so its weight normalized by this minimum distance is $w/2^{n-k} = 2^k p_i = \lambda$. We reduce A_i itself, rather than A_i^2, whose real degree could be $2k$.

We use the Kasami–Tokura theorem on Reed–Muller codewords of weight less than twice the minimum distance (references below). For $k \ge 2$, any nonzero word of $\mathrm{RM}(k, n)$ with weight strictly below $2 \cdot 2^{n-k}$ has normalized weight

$$\lambda = 2 - 2^{-j}, \quad \text{for an integer } j \ge 0. \tag{7}$$

Since $w < 2 \cdot 2^{n-k}$ is the same as $\lambda < 2$, (7) applies exactly when $\lambda < 2$. The theorem also says which j occur for given n and k. We use only the fact that λ has the form (7), so we allow every $j \ge 0$. As above, we have $n \ge k + 2$. The theorem applies to words of algebraic degree at most k. A nonzero word of algebraic degree at most $k - 1$ lies in $\mathrm{RM}(k-1, n)$, so its weight is at least 2^{n-k+1}, twice the minimum distance of $\mathrm{RM}(k, n)$. The theorem is due to Kasami–Tokura (1970), with Berlekamp–Sloane, DOI 10.1109/TIT.1970.1054545. A published explicit statement and the two normal forms appear in Carlet (2024), section 2, p.10613, `https://www.aimspress.com/aimspress-data/math/2024/5/PDF/math-09-05-518.pdf` .

For $k \ge 2$, combine (6) and (7). By (6), $M\lambda$ is an even integer, and by (7), $M\lambda = M(2 - 2^{-j}) = 2^{k+1} - 2^{k-j}$. For $j \le k - 1$ this is even. For $j \ge k$ it is either nonintegral ($j > k$) or odd ($j = k$). When $k = 1$ ($d = 2$), (6) says λ is a positive integer, so $\lambda < 2$ forces $\lambda = 1$, which is (8) below with $k = 1$. (The certificate in Section 5 starts at $d = 4$, so this case is recorded only for completeness.) Hence, for every $k \ge 1$, every possible value of λ below 2 lies in the finite set

$$\{2 - 2^{-j} : j = 0, \ldots, k-1\}. \tag{8}$$

The first terms are $1, 3/2, 7/4, 15/8, 31/16, \ldots$. For $\lambda \ge 2$ we use only the grid (6). No further information about Reed–Muller weights in $[2, 2.5)$ (Kasami–Tokura–Azumi) is used.[1]

[1]The attribution to third and higher below-two weights is open. An AI referee's ablation finds that, given the grid (6), excluding normalized weights in $(1, 3/2)$ alone already gives the same rational, while the ablation numbers in Section 5 disagree. The bound does not depend on that attribution. See Correction 2, printed at the end of Section 5.

5 Finite scalar optimization, with the floor treated correctly

Suppose c bounds W_{d-1}. To exclude $N \geq n$, (5) puts λ in $[1, t]$, where $t = dM/n$. Since the set of possible λ for $N \geq n$ is contained in this interval, it suffices to show every allowed λ in $[1, t]$ has $F_d(c, \lambda) < n$. This excludes all $N \geq n$ at once, not merely $N = n$.

Use $c \in [1/2, (d-1)/2]$, checked inductively in the certificate. Within a cell $h \leq 2k\lambda < h + 1$, the value (4) is

$$2^d c + h + (1 - 2c)\lambda,$$

which is nonincreasing in λ. It is NOT safe to substitute t directly inside its floor. Below 2 we test every member of (8) not exceeding t. For $\lambda \geq 2$, the possible cell indices are $4k$ through $\lfloor 2kt \rfloor$. On the grid (6), the first value in cell h is

$$\lambda_h = \frac{2}{M} \left\lceil \frac{hM}{4k} \right\rceil . \tag{9}$$

Discard it if $\lambda_h > t$ or $2k\lambda_h \geq h + 1$. Every other grid value in that cell has no larger budget. Thus (8),(9) form a complete finite cover of all possible cases. Every such budget is strictly below n in `caps.json`. The resulting $c_d = (n-1)/2^d$ is valid by (4),(5) and the CHS identity.

Starting with $c_1 = 1/2, c_2 = 1, c_3 = 5/4$, proceed through $d = 96$. The certificate uses each already established c_{d-1} before deriving c_d. There is no circular use of the eventual constant. Some initial ceilings are:

d	R_d ceiling	c_d
4	24	3/2
5	54	27/16
6	119	119/64
7	254	127/64
8	528	33/16
9	1083	1083/512
10	2199	2199/1024
11	4439	4439/2048
12	8928	279/128

The frozen packet stores the most dangerous finite case at each first excluded count, its complete case count, and a feasible scalar point at the preceding count. Checker B independently enumerates all floor cells; checker A uses an independently derived collapsed envelope. For the latter, write $a = 2c - 1 \in [0, k-1]$. Consecutive first grid values in occupied cells $h < h'$ differ by at most $(h' - h)/(2k) + 2/M$. Therefore their maximal budgets increase by at least

$$\begin{aligned} &(h' - h)\left(1 - \frac{a}{2k}\right) - \frac{2a}{M} \\ &\geq 1 - \frac{a}{2k} - \frac{2a}{M} > 0. \end{aligned}$$

The last strict inequality is checked exactly for every degree; universally it follows for $k \geq 3$ from $(k-1)/(2k) + 2(k-1)/2^k < 1$ (at $k = 3$ this is $5/6$; at $k \geq 4$ the first term is below $1/2$ and

the second is at most 3/8). Thus the rightmost occupied high cell suffices for checker A. It locates that cell from the last grid point below t and tests its first grid point. Checker B uses none of this envelope shortcut.

Two additional certified variants quantify the inputs. Dropping the RM gaps altogether and allowing a continuum $\lambda \geq 1$ gives a bound below 2.2952887883. Retaining only $\lambda = 1$ or $\lambda \geq 3/2$ gives a bound below 2.2108998048. The third and higher below-two weights give the displayed 2.1974348818 bound. Each variant has its own inductive caps and both checkers verify them. These are candidate, unrefereed consequences of the same written lemmas, not claims about the exact Boolean maxima.

Correction: The attribution to third and higher below-two weights is open. The referee's ablation finds that, given the energy grid, the single fact that no normalized weight lies in $(1, 3/2)$ already gives the same rational. Section 5's ablation numbers disagree, and this has not been resolved; the bound does not depend on that attribution. This is correction 2 in `CORRECTIONS.md`.

6 An elementary infinite tail

We need a global bound, not just a horizon calculation. The CHS/Shinkar–Tal lower family starts with $\Xi_1(x) = x$ and, on disjoint bit variables, uses

$$\Xi_d = s + (1 - s - t)\Xi_{d-1}(x) + (t - s)\Xi_{d-1}(y).$$

For $(s, t) = (0, 0), (0, 1), (1, 0), (1, 1)$ the output is respectively $\Xi_{d-1}(x)$, $\Xi_{d-1}(y)$, $1 - \Xi_{d-1}(y)$, $1 - \Xi_{d-1}(x)$, so induction proves Booleanity. Its degree is d: the formula has degree at most d, and a degree-$(d-1)$ monomial in the first copy produces a noncanceling degree-d term containing s. Each input copy remains relevant by selecting it. Both selectors remain relevant: choosing both copy outputs zero makes the result s, while choosing the first zero and the second one makes it t. Thus its count L_d satisfies $L_1 = 1$ and $L_d = 2L_{d-1} + 2$. Solving this recursion gives

$$L_d = 3 \cdot 2^{d-1} - 2 = 3M - 2.$$

Thus $R_d \geq 3M - 2 \geq 2M$ for $d \geq 2$. For the least-influence coordinate of its count extremizer, (5) now yields $\lambda \leq d/2$. Apply (4) with the ACTUAL $c = W_{d-1}$, discard its floor downward, and divide by 2^d. Since $1/2 \leq W_{d-1} \leq (d-1)/2$, the multiplier $2d - 1 - 2W_{d-1}$ is positive and at most $2d - 2$. Hence

$$\begin{aligned} W_d &\leq W_{d-1} + \frac{\lambda(2d - 1 - 2W_{d-1})}{2^d} \\ &\leq W_{d-1} + \frac{d(d-1)}{2^d}. \end{aligned} \tag{10}$$

No block-sensitivity theorem or inherited numerical block certificate is required for (10). For $H \geq 1$, write $d = H + j$ and sum $j \geq 1$. The elementary geometric-series identities are

$$\sum 2^{-j} = 1, \qquad \sum j 2^{-j} = 2, \qquad \sum j^2 2^{-j} = 6.$$

The first follows by summing the geometric series; applying $x\,d/dx$ to $\sum_{j\ge1} x^j = x/(1-x)$ once and twice, then setting $x = 1/2$, gives the other two. Expansion of $(H+j)(H+j-1)$ gives

$$\sum_{d>H} \frac{d(d-1)}{2^d} = \frac{H^2+3H+4}{2^H}. \tag{11}$$

The computed caps are nondecreasing through H. Every smaller degree is therefore bounded by c_H; for larger degrees telescope (10) and apply (11). Consequently $C_{44} = \sup_d R_d/2^d$ satisfies

$$\begin{aligned} C_{44} &\le c_{96} + \frac{9508}{2^{96}} \\ &= \frac{1740987279236787915969864029 33}{7922816251426433759354395 0336} \\ &< 2.1974348818. \end{aligned} \tag{12}$$

The exact improvement over round five's frozen rational is

$$\frac{76852852570035861934916069377}{79228162514264337593543950336} > \frac{1}{1000}.$$

This is an exact rational comparison. Rounding displayed decimals never enters the proof.

7 What the recurrence and the scalar limit do and do not establish

The inherited valid block recurrence is

$$\begin{aligned} &U(0,d) = 0, \\ &U(b,d) = \min\left(c_d, \frac{d}{2^d} + \max_{1\le j\le d} U(b-1,j)\right), \\ &U(b,d) = U(B_d,d) \quad \text{for } b > B_d. \end{aligned}$$

Its proof and raw block duals are retained in `baseline/PROOF.md`, sections 5–6. In outline, restricting a maximum monomial of size d costs $d/2^d$ in (3) and lowers block sensitivity by one: the nonzero full monomial coefficient guarantees nonconstancy on its coordinates at every outside input; this gives one sensitive block disjoint from all sensitive blocks of the restricted function. Induct on b and take the maximum over every degree $j \le d$. Freezing a column above B_d is valid because that class of functions cannot grow further.

Checker B rechecks every retained Épi moment certificate and all 37 Fourier-lattice duals before re-running this recurrence with the inherited B_d. Checker A separately computes integer recurrence rows using only $B_d = d^2$, a consequence of the external Wellens theorem $3\,\mathrm{bs}(\mathrm{bs}-1) \le 2(d^4-d^2)$, Theorem 4.1, equation (4.3), `https://arxiv.org/html/2005.00566` . Both reach exactly c_d in EVERY final column through degree 96. Thus the current block inputs are inactive at the saturation rows of this run. The accepted global bound (12) has the stronger elementary tail (11) and does not rely on those block premises. The saturation thresholds and slack are in `analysis.json`. A smaller B_d can help only if it crosses below its saturation threshold, with other inputs fixed; a decrement that leaves it at or above that threshold has zero effect in this recurrence.

There is also a precise limit for the scalar relaxation just specified. Define its next count as the largest integer N for which (4),(5),(6),(8) are jointly feasible, allowing every high grid value ≥ 2. The packet's feasible scalar points at its count ceilings and the strict exclusions at the next integers prove that the computed counts are exactly optimal for THIS relaxation. Those scalar points are not Boolean existence witnesses.

Let c be the previous scalar cap. Its denominator divides 2^{d-1}, so $N_0 = 2^d c$ is an integer. The case $\lambda = 1$ has budget $N_0 + 2d - 1 - 2c \geq N_0$ and satisfies $N_0\lambda \leq dM$ whenever $c \leq d/2$. It belongs to every allowed low spectrum. Hence the next cap is at least c. Every feasible scalar point at the largest count also has $N \geq 3M - 2$, so the same estimates as in (10) bound the scalar increment by $d(d-1)/2^d$. The maintained upper condition $c_d \leq d/2$ continues after $d = 96$ by induction, since this increment is less than $1/2$ for $d \geq 6$. Therefore the infinite scalar sequence exists, is nondecreasing, and its limit lies in

$$\left[\frac{174098727923678791596986393425}{2^{96}},\ \frac{174098727923678791596986402933}{2^{96}}\right]. \tag{13}$$

This pins its limit to $2.197434881723198\ldots$ with interval width $9508/2^{96} < 1.3 \cdot 10^{-25}$. It quantifies the end of this particular energy and weight-spectrum relaxation. It is not an optimality theorem for all uses of Reed–Muller structure. Full ternary real-degree compatibility, sharper control sets for actual higher-weight derivatives, correlations between derivatives, or new block bounds crossing the saturation thresholds could improve it. The known lower bound is $3/2$ from the family above, leaving a candidate gap about 0.6974348817. The remaining gap cannot be removed by merely iterating this already saturated scalar relaxation farther.

Related energy prior art: Potapov, *On the number of relevant variables for discrete functions*, Cryptography and Communications 17 (2025), 989–998, `https://link.springer.com/article/10.1007/s12095-025-00797-4` , section 5, also uses Fourier energy and changed edges to bound ternary relevance. Its root-of-unity output encoding can double the real degree of a ternary polynomial in our encoding. We credit the related method; no novelty is claimed for the elementary real-energy argument in section 2.

Corrections

Corrections to PROOF.md

PROOF.md is frozen so that its hash matches the receipts. A referee found two wording defects. Neither changes the mathematics.

1. Section 3, "Applying them successively to coordinates in H, or equivalently inducting on |H|". Read literally, this charges each later coordinate at its weight after restriction, and that inequality runs the wrong way. The correct reading: iterate over H for each coordinate outside H, then charge H at its original weights.

2. Sections 4 and 5 credit the third and higher below-two weights for the final bound. The referee's ablation finds that, given the energy grid, the single fact that no normalized weight lies in (1, 3/2) already gives the same rational. Section 5's ablation numbers disagree with that, and this hasn't been resolved. Treat the attribution as open. The bound doesn't depend on it.

External premises are R3 = 10 and the second Reed-Muller weight 3/2. The Kasami-Tokura-Azumi 1976 results aren't a premise. The claim in KTA_AUDIT.md about a table omission is unverified.

External premises and cited results

The external premises identified in CORRECTIONS.md are $R_3 = 10$ and the second Reed–Muller weight 3/2 (no normalized weight lies in $(1, 3/2)$). The degree-three result is cited through Tarannikov–Kirienko, https://eprint.iacr.org/2000/050, and Krotov–Valyuzhenich, https://arxiv.org/html/2311.05566v2, Section 3. These results are cited, not reproved.

Section 4 cites the fuller below-two Kasami–Tokura spectrum, crediting Berlekamp–Sloane for low binary weights: Kasami–Tokura (1970), https://doi.org/10.1109/TIT.1970.1054545; see also Carlet (2024), Section 2, p. 10613, linked there. Correction 2 leaves the need for the higher below-two weights for this bound open.

Section 1 proves and cites the Chiarelli–Hatami–Saks identity. Section 7 cites Wellens's recurrence and block theorem, whose premises are not needed for (12). Valyuzhenich's minimum-support classification is context, rather than a premise of Section 2's energy inequality.

KTA audit caveat. The Kasami–Tokura–Azumi (1976) results are not a premise. KTA_AUDIT.md reports that its original PDF endpoint and the Elsevier API denied access, and that the full 1974 preprint was not reconstructed. Its claim about an omission in the final table is *unverified*, as CORRECTIONS.md explicitly states. The proof allows all energy-grid values at and above 2 and uses no KTA exclusions in the range 2 to 2.5. The audit's claimed higher-weight numerical gain is also subject to correction 2.

Certificates and reproduction

Pinned source folder:

https://github.com/rpinho/math-certificates/tree/09c84234137d3fadee1620dbc1c2ca58d2756ce2/c44/reed-muller-caps

The frozen PROOF.md SHA-256, checked against MANIFEST.json, is

af599ddff4732631a1c0565d5a5b3a90510d53955436b085b55883d85dcc4c47

The packet includes the following files and subfolders.

- Written sources: `PROOF.md`, `CORRECTIONS.md`, `README.md`, `PRIOR_ART.md`, `KTA_AUDIT.md`, and `review/`.
- Exact inputs and generation: `caps.json`, `candidate.json`, `generate.py`, `analyze.py`, `analysis.json`, and `analysis.log`.
- Checkers: `verify_caps_a.py` and `verify_caps_b.py`. Checker B also reads `baseline/epi_certificate.json` and `baseline/spectral_certificate_round3.json`; `baseline/` retains the inherited proofs and certificates.
- Receipts and hashes: `check_a.log`, `check_b.log`, `verification_a.json`, `verification_b.json`, `FINAL_RECEIPT.json`, `MANIFEST.json`, `SHA256SUMS`, and `PROCESS_LEDGER.json`.
- Finite controls: `control_boolean.cpp`, `control_ternary.cpp`, `control_corruption.py`, `control_corruption.json`, `boolean.log`, `ternary.log`, and `corruption.log`. `jobs/` retains the original reproduction wrappers.

From a writable copy of the pinned folder, using Python 3.10 or later (standard library), the two checker commands are

```
nice -n 10 python3 -B verify_caps_a.py candidate.json
nice -n 10 python3 -B verify_caps_b.py candidate.json
```

The scripts write `verification_a.json` and `verification_b.json` beside themselves. Both were replayed for this PDF and returned PASS with the rational in (12). They check arithmetic conditional on the written lemmas and external premises, not those universal statements themselves.

Credit and later work

Thanks to Alexandr Valyuzhenich for suggesting the $\{0, 1\}$ convention, which shortens Section 4 without changing its conclusions.

Following the source README and prior-art note, credit goes to Chiarelli, Hatami and Saks for the weighted framework (including the corrected duplication in Chiarelli's thesis); Wellens for the recurrence; Berlekamp and Sloane, Kasami, Tokura and Azumi for the weight spectra; Valyuzhenich for minimum-influence structure; BakeLab/Épi for inherited exact polynomial certificates; and Shinkar and Tal for the lower family also credited in Section 6. The degree-three citations are due to Tarannikov and Kirienko, and Krotov and Valyuzhenich. Potapov's related ternary spectral-energy method is credited in Section 7. The source prior-art note and KTA audit also discuss Carlet, Rolland, Klein and Storme, Krotov's public discussion, Lou and Wang, and Wei et al.; these contextual references do not supply additional premises for (12). No novelty, priority, or optimality is claimed.

Later work in the same repository (`small-degree-classification`) gives $C_{44} < 1.9916994577$, also unrefereed; this PDF concerns the requested 2.1974348818 proof.